Sprint Plan & Definition of Done (DoD)

Project: Smart Irrigation Alert Dashboard
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Sprint Plan Breakdown

Sprint 1: Foundation, Architecture & SCM (Days 1–5)

- **Goal:** Define problem scope, setup Git repository, configure Maven Spring Boot skeleton, and initialize architecture.
- **Tasks Included:** Task 01 to 04 (Scope, Planning, Architecture, Git/GitHub Init).
- **Deliverables:** Working project repository, `.gitignore`, initial commit, architecture documentation.

Sprint 2: Core Development & Quality Assurance (Days 6–10)

- **Goal:** Implement the 5 MVP features, simulate sensor feeds, resolve merge conflicts, and implement automated tests.
- **Tasks Included:** Task 05, 06, 09, 10 (Branching, MVP Completion, Selenium, Jenkins Quality Gate).
- **Deliverables:** Fully functioning dashboard at localhost:8080, passing Selenium tests, Git tag `v1.0`.

Sprint 3: CI/CD Pipeline & Containerization (Days 11–15)

- **Goal:** Automate build and deployment pipelines using Jenkins and Docker.
- **Tasks Included:** Task 07, 08, 11, 12 (Jenkins Freestyle, Pipeline, Docker Image, Docker Hub CD).
- **Deliverables:** Automated Jenkins Pipeline stage view, versioned Docker image.

Sprint 4: Provisioning, Reliability & Final Release (Days 16–20)

- **Goal:** Write Ansible configuration playbooks, validate idempotency, verify rollback, and compile reports.
- **Tasks Included:** Task 13 to 15 (Ansible Config, Provisioning/Rollback, Final Docs).
- **Deliverables:** Ansible `setup.yml`, idempotency logs, final project PDF/MD report, Viva slide deck.

Definition of Done (DoD) Checklist

A feature is considered "Done" ONLY when it satisfies all of the following:

- ✓ **Code Completeness:** Code follows clean standards and all 5 MVP criteria are verified.

✓ **Build & Compilation:** Compiles without errors via `mvn clean compile` and packages into a `.jar`.

✓ **Automated Testing:** All Selenium test cases pass with 100% green status. No regressions.

✓ **Version Control:** Committed with descriptive messages, merged to main, tagged `v1.0`.

✓ **Containerization:** Dockerfile builds clean image. Jenkins pipeline stages execute successfully.

✓ **Documentation:** All logs/screenshots saved in folders, `CHECKPOINTS.md` marked done.